Assignment 6 – DDL

Car Rental Agency

1. **Customer:** Nevo Alani
2. **Designer:** Ofek Goldstein
3. **Reviewer:** Yaron Besser

Client - "FastTrack Tolls"

A national toll road management company

A national toll road operator manages a central intercity highway. When a vehicle enters a toll road it passes through an intersection and the system records the entry. Upon exiting at a second intersection the system calculates the applicable fee based on the route, the vehicle classification, and the current rate table, then records the completed drive. Registered customers prepay or pay monthly: all drives made by their vehicles during a billing period are grouped into a single invoice and sent at the end of the period. The company needs to track roads and their interchanges, the drive history, the customers who own vehicles, and the invoices that are generated for billing.

Requirements

4. The system should store **customer information** including full name, contact email, phone number, billing address, and registration date.
5. The system should store **vehicle information** including license plate, vehicle type (motorcycle, private car, light truck, heavy truck, or bus), make, model, color, and the owning customer.
    - A single customer may own multiple vehicles.
6. The system should store **intersections** as the entry and exit points of each road. Each intersection must record its name and its position along the road (1-8).
7. The system should store **toll rates** as the fee charged for driving one section, differentiated by vehicle type. Rates must be versioned with a validity date range so that historical billing is never affected by future rate changes.
8. The system should record **every drive**: which vehicle made it, which intersection it entered at and which it exited at, the timestamps of entry and exit, the total number of sections, and the toll amount charged.

Design

Discussion – Rate Versioning

Customer: The toll rates change depending on the vehicle type, but more importantly, we need to ensure that historical billing is never affected by future rate changes. If a customer disputes a bill from six months ago, the system must reflect the exact rates that were active on that specific date.

Designer: I agreed with the customer's requirement to version the rates. I explained that if we simply update a standard price column when rates increase, all past drives tied to that rate would suddenly appear more expensive. To prevent this, we need to implement rate versioning, ensuring each specific rate entry will have a validity date range

Final Decision: We agreed to implement rate versioning by storing toll rates with a validity date range so that historical billing is never affected by future rate changes.

System Overview

FastTrack Tolls operates a single major toll highway with exactly eight numbered interchanges (Intersections, positions 1–8). The core billing unit is a Drive: a vehicle enters at one interchange and exits at another, and the fee is proportional to the number of sections travelled (one section = the segment between two consecutive interchanges). Because vehicle type determines the persection rate, the chargeable event belongs to the vehicle, not directly to the customer.

Billing is customer-level and monthly: all drives made by any vehicle owned by a given customer in the billing period are aggregated into a single Invoice. This means the customer is the billing entity, the vehicle is the operational entity, and the two must be stored separately.

Toll rates change over time. The system must always be able to reproduce the exact charge that was in force on the date of any historical drive, regardless of subsequent rate changes. This requires rate versioning.

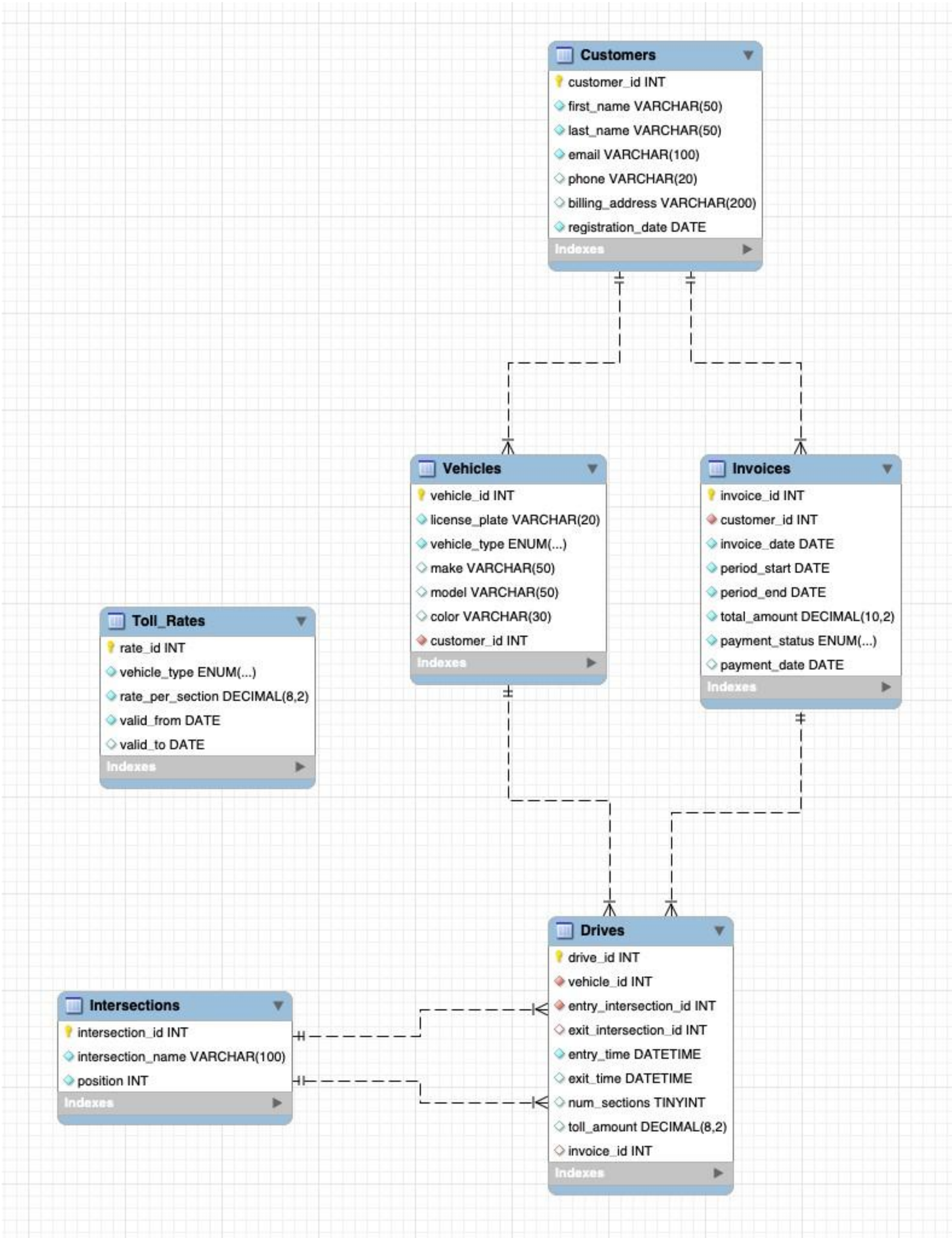
The database therefore needs six tables (explaination per table is in the code): **Customers, Vehicles, Intersections, Toll_Rates, Invoices, and Drives.**

Normalization

The schema fully satisfies **3NF**:

- **Structure:** All tables use a single-column primary key, and all non-key attributes depend on it.
- **Separation:** Customers and Vehicles are separated to prevent customer data redundancy and 2NF update anomalies.
- **Deviation:** Instead of computing fees on the fly, toll_amount is snapshotted into Drives at exit time. While technically derivable from Toll_Rates, storing it freezes historical billing and preserves audit integrity against future price changes.

Scheme



Customers
customer_id INT
first_name VARCHAR(50)
last_name VARCHAR(50)
email VARCHAR(100)
phone VARCHAR(20)
billing_address VARCHAR(200)
registration_date DATE
Indexes
Vehicles
vehicle_id INT
license_plate VARCHAR(20)
vehicle_type ENUM(...)
make VARCHAR(50)
model VARCHAR(50)
color VARCHAR(30)
customer_id INT
Indexes
Invoices
invoice_id INT
customer_id INT
invoice_date DATE
period_start DATE
period_end DATE
total_amount DECIMAL(10,2)
payment_status ENUM(...)
payment_date DATE
Indexes
Toll_Rates
rate_id INT
vehicle_type ENUM(...)
rate_per_section DECIMAL(8,2)
valid_from DATE
valid_to DATE
Indexes
Drives
drive_id INT
vehicle_id INT
entry_intersection_id INT
exit_intersection_id INT
entry_time DATETIME
exit_time DATETIME
num_sections TINYINT
toll_amount DECIMAL(8,2)
invoice_id INT
Indexes
Intersections
intersection_id INT
intersection_name VARCHAR(100)
position INT
Indexes

SQL Code

```sql
-- ============================================================
-- FastTrack Tolls – Toll Road Management System
-- Assignment 6 – DDL
-- ============================================================
CREATE SCHEMA IF NOT EXISTS fasttrack_tolls;
USE fasttrack_tolls;

-- 1. Customers
-- • email: UNIQUE business key for login and contact.
-- • customer_id: Surrogate PK prevents foreign key breakage if email/address changes.
CREATE TABLE Customers (
    customer_id       INT          NOT NULL AUTO_INCREMENT,
    first_name        VARCHAR(50)  NOT NULL,
    last_name         VARCHAR(50)  NOT NULL,
    email             VARCHAR(100) NOT NULL,
    phone             VARCHAR(20),
    billing_address   VARCHAR(200),
    registration_date DATE         NOT NULL,
    CONSTRAINT pk_customers       PRIMARY KEY (customer_id),
    CONSTRAINT uq_customers_email UNIQUE      (email)
);

-- 2. Vehicles
-- • vehicle_type: ENUM enforces the 5 valid domain classifications.
-- • license_plate: UNIQUE to prevent legal duplicates.
-- • Separated from Customers to avoid data redundancy.
CREATE TABLE Vehicles (
    vehicle_id    INT         NOT NULL AUTO_INCREMENT,
    license_plate VARCHAR(20) NOT NULL,
    vehicle_type  ENUM('motorcycle','private_car',
                       'light_truck','heavy_truck','bus') NOT NULL,
    make          VARCHAR(50),
    model         VARCHAR(50),
    color         VARCHAR(30),
    customer_id   INT         NOT NULL,
    CONSTRAINT pk_vehicles      PRIMARY KEY (vehicle_id),
    CONSTRAINT uq_vehicles_lp   UNIQUE      (license_plate),
    CONSTRAINT fk_vehicles_cust FOREIGN KEY (customer_id)
        REFERENCES Customers (customer_id)
);

-- 3. Intersections
-- • position: Constrained to 1–8 via CHECK and UNIQUE constraints.
-- • Distinct entity allows stable naming without updating thousands of Drives.
-- • No road_id FK needed as the system manages a single road.
CREATE TABLE Intersections (
    intersection_id   INT          NOT NULL AUTO_INCREMENT,
    intersection_name VARCHAR(100) NOT NULL,
    position          INT          NOT NULL,
    CONSTRAINT pk_intersections  PRIMARY KEY (intersection_id),
    CONSTRAINT uq_intersect_pos  UNIQUE      (position),
    CONSTRAINT chk_intersect_pos CHECK       (position BETWEEN 1 AND 8)
);

-- 4. Toll_Rates
-- • rate_per_section: Base price for one segment per vehicle_type.
-- • Versioning: valid_from/valid_to dates track historical rates (NULL valid_to = active).
-- • Per-section model chosen over 56 entry-exit pairs for easier auditing and maintenance.
CREATE TABLE Toll_Rates (
    rate_id          INT          NOT NULL AUTO_INCREMENT,
    vehicle_type     ENUM('motorcycle','private_car',
                          'light_truck','heavy_truck','bus') NOT NULL,
    rate_per_section DECIMAL(8,2) NOT NULL,
    valid_from       DATE         NOT NULL,
    valid_to         DATE,        -- NULL = currently active
    CONSTRAINT pk_toll_rates PRIMARY KEY (rate_id)
);
```

```sql
-- 5. Invoices
--  • References Customers (not Vehicles) to group all owned vehicles into one bill.
--  • total_amount: Stored snapshot of billed amount to freeze financial history, preventing drift.
CREATE TABLE Invoices (
    invoice_id     INT           NOT NULL AUTO_INCREMENT,
    customer_id    INT           NOT NULL,
    invoice_date   DATE          NOT NULL,
    period_start   DATE          NOT NULL,
    period_end     DATE          NOT NULL,
    total_amount   DECIMAL(10,2) NOT NULL,
    payment_status ENUM('pending','paid','overdue','cancelled')
                                 NOT NULL DEFAULT 'pending',
    payment_date   DATE,
    CONSTRAINT pk_invoices     PRIMARY KEY (invoice_id),
    CONSTRAINT fk_invoice_cust FOREIGN KEY (customer_id)
        REFERENCES Customers (customer_id)
);

-- 6. Drives
--  • exit_intersection_id/exit_time: NULL while drive is in progress.
--  • num_sections: Denormalized (|exit - entry|) to optimize queries and handle exceptions.
--  • toll_amount: Snapshot of calculated fee at exit to freeze historical pricing.
--  • invoice_id: NULL until the monthly billing run processes the drive.
--  • References Vehicles (chargeable event), not Customers.
CREATE TABLE Drives (
    drive_id             INT           NOT NULL AUTO_INCREMENT,
    vehicle_id           INT           NOT NULL,
    entry_intersection_id INT          NOT NULL,
    exit_intersection_id  INT,
    entry_time           DATETIME      NOT NULL,
    exit_time            DATETIME,
    num_sections         TINYINT,
    toll_amount          DECIMAL(8,2),
    invoice_id           INT,
    CONSTRAINT pk_drives        PRIMARY KEY (drive_id),
    CONSTRAINT fk_drive_vehicle FOREIGN KEY (vehicle_id)
        REFERENCES Vehicles      (vehicle_id),
    CONSTRAINT fk_drive_entry   FOREIGN KEY (entry_intersection_id)
        REFERENCES Intersections (intersection_id),
    CONSTRAINT fk_drive_exit    FOREIGN KEY (exit_intersection_id)
        REFERENCES Intersections (intersection_id),
    CONSTRAINT fk_drive_invoice FOREIGN KEY (invoice_id)
        REFERENCES Invoices      (invoice_id)
);
```

Review

I confirm that the finalized FastTrack Tolls database effectively meets all customer requirements while adhering to Third Normal Form (3NF) principles. Key architectural decisions-such as isolating customer and vehicle data to prevent redundancy, implementing date ranges for toll rates to protect historical billing, and storing calculated totals on drives-ensure absolute data integrity. Ultimately, this design delivers a robust, scalable foundation for tracking road usage and generating accurate billing.